Architect’s Brief

Clinic Appointment & Intake System

1. Architecture Purpose

This brief prepares the Clinic Appointment & Intake System for system architecture modeling. It identifies the likely system boundary, external participants, major internal responsibilities, key integrations, data ownership concerns, workflows, and quality drivers.

This brief is not a product requirements document and does not define implementation design or code structure.

2. System Boundary

The subject system is the **Clinic Appointment & Intake System**.

The system owns the digital scheduling and intake experience, appointment workflow coordination, reminder orchestration, intake persistence, staff review experience, integration status tracking, and synchronization coordination with external systems.

The system does not own the complete clinical record, clinical documentation, provider credentialing, billing, claims processing, or the external electronic health record system.

3. External Participants

Participant	Role in the Architecture
Patient	Initiates scheduling, rescheduling, cancellation, and intake submission workflows.
Clinic Staff	Reviews appointments, manages appointment status, monitors intake completion, and resolves operational issues.
Provider	Reviews submitted intake information before the visit.
Clinic Administrator	Configures appointment types, intake requirements, reminder rules, and access policies.
Electronic Health Record System	External clinical system that receives appointment and intake data and may provide reference identifiers.
Notification Provider	External service used to send email and SMS messages.

Participant	Role in the Architecture
Insurance Eligibility Service	Optional external service for eligibility checks.
Identity Provider	Optional external authentication and role source for staff and providers.

4. Candidate Internal System Responsibilities

Responsibility Area	Description
Patient Scheduling Experience	Patient-facing workflow for slot search, appointment request, booking, rescheduling, and cancellation.
Staff Appointment Management	Staff-facing workflow for reviewing, confirming, rejecting, rescheduling, and canceling appointments.
Appointment Coordination	Applies scheduling rules, manages appointment state, and coordinates downstream effects.
Intake Form Management	Manages intake form definitions, drafts, validation, submission, review status, and correction requests.
Reminder Coordination	Determines when confirmations and reminders should be sent and tracks notification status.
EHR Synchronization	Coordinates outbound appointment and intake synchronization and records sync outcomes.
Eligibility Coordination	Coordinates optional eligibility requests and records eligibility results.
Access Control	Enforces role-based access for staff, providers, administrators, and patient workflows.
Audit Logging	Records important lifecycle events and user/system actions.
Integration Monitoring	Tracks failed, pending, retried, and completed external integration attempts.

5. Candidate Data Stores

Data Store	Architectural Purpose
Appointment Store	Stores appointment records, appointment status, scheduling metadata, and related

Data Store	Architectural Purpose
	external identifiers.
Intake Store	Stores intake form definitions, drafts, submissions, and review status.
Configuration Store	Stores clinic-specific appointment types, cancellation rules, reminder rules, intake requirements, and feature settings.
Notification Queue	Buffers confirmation and reminder requests before delivery.
Integration Status Store	Tracks synchronization attempts, external references, retry state, and failure details.
Audit Log Store	Stores significant user and system events for traceability.
Eligibility Result Store	Stores eligibility request metadata and result summaries where eligibility checking is enabled.

6. Candidate External Interfaces

Interface	Provider	Consumer	Notes
Patient Scheduling Interface	Clinic Appointment & Intake System	Patient-facing experience	Supports slot search, booking, rescheduling, cancellation, and appointment details.
Staff Appointment Interface	Clinic Appointment & Intake System	Staff-facing experience	Supports appointment review, status changes, overrides, and operational search.
Intake Submission Interface	Clinic Appointment & Intake System	Patient-facing experience	Supports intake draft save, validation, submission, and correction.
Intake Review Interface	Clinic Appointment & Intake System	Staff/provider experience	Supports intake review, missing information visibility, and review status.
EHR Synchronization Interface	Electronic Health Record System	Clinic Appointment & Intake System	Supports appointment and intake synchronization.

Interface	Provider	Consumer	Notes
			External API shape will depend on the EHR.
Notification Delivery Interface	Notification Provider	Clinic Appointment & Intake System	Supports email and SMS confirmations and reminders.
Eligibility Check Interface	Insurance Eligibility Service	Clinic Appointment & Intake System	Optional integration for insurance eligibility requests.
Staff Authentication Interface	Identity Provider	Clinic Appointment & Intake System	Optional integration for staff and provider identity.

7. Primary Workflows for Architecture Modeling

Workflow	Summary
Schedule Appointment	Patient searches availability, selects a slot, submits booking/request, receives confirmation, and appointment is synchronized externally.
Reschedule Appointment	Patient or staff changes appointment date/time; system updates appointment state, sends notification, and synchronizes change.
Cancel Appointment	Patient or staff cancels appointment; system validates rules, updates status, sends notification, and synchronizes cancellation.
Submit Intake	Patient completes forms, system validates required fields, stores submission, and makes it available for review.
Review Intake	Staff or provider reviews submitted intake and marks review state or requests corrections.
Send Reminder	System identifies reminder candidates, prepares message, sends through notification provider, and records delivery status.
Sync Appointment to EHR	System sends appointment data, records external identifiers, handles success/failure, and retries where

Workflow	Summary
	appropriate.
Sync Intake to EHR	System sends submitted intake data, records outcome, and exposes failures to staff.
Check Eligibility	System submits eligibility request, records response, and displays status to authorized staff.

8. Key State Models

Appointment State

Candidate states:

1. Requested.
2. Confirmed.
3. Rescheduled.
4. Canceled.
5. Completed.
6. No-show.
7. Sync pending.
8. Sync failed.

The architecture should clarify whether synchronization state is part of appointment state or tracked separately.

Intake State

Candidate states:

1. Not started.
2. In progress.
3. Submitted.
4. Needs correction.
5. Reviewed.
6. Sync pending.
7. Sync failed.

Integration Attempt State

Candidate states:

1. Pending.
2. Sent.
3. Succeeded.

4. Failed.
5. Retrying.
6. Abandoned.
7. Requires manual review.

9. Key Quality Drivers

Quality Driver	Architectural Implication
Privacy	Patient information should be minimized, access-controlled, encrypted, and protected in logs and notifications.
Auditability	Appointment changes, intake submissions, reviews, reminders, and external sync attempts require traceable event history.
Reliability	Failed external integrations should not lose state; retry and manual review paths are needed.
Availability	Patient scheduling and intake should remain available outside clinic hours, even if some integrations are temporarily unavailable.
Data Consistency	Appointment and intake status must remain understandable across local state and external system synchronization state.
Usability	Patient workflows should remain simple despite identity verification, required forms, and validation rules.
Configurability	Clinics need control over appointment types, reminders, intake requirements, and cancellation policies.
Security	Staff/provider access and patient access require distinct authentication and authorization patterns.

10. Architecture Risks

Risk	Architectural Concern
EHR API variability	Different electronic health record systems may require adapter-specific synchronization behavior.
Ambiguous source of truth	Appointment, patient, and intake data ownership must be defined to avoid

Risk	Architectural Concern
	inconsistent updates.
Partial external failures	Appointment creation may succeed locally while EHR sync or notification delivery fails.
Sensitive notification content	Reminder content must avoid exposing protected or sensitive health details.
Intake versioning	Submitted intake may need versioning if patients revise information after submission.
Identity matching	Matching patient-entered data to EHR patient records may be error-prone.
Eligibility dependency	Eligibility service downtime should not necessarily block scheduling.
Role complexity	Staff, providers, and administrators may require different access to the same appointment and intake records.

11. Open Architecture Questions

1. Should patients authenticate with a full account, magic link, verification code, or clinic-provided token?
2. Is appointment availability mastered locally, in the electronic health record system, or in another scheduling system?
3. Is this system allowed to directly confirm appointments, or must staff approve all appointment requests?
4. Which appointment changes must be synchronized immediately versus queued?
5. What data should be sent to the electronic health record system for each intake submission?
6. Should intake drafts expire after a configured period?
7. Should patients be able to update submitted intake forms after staff review?
8. What happens when a patient submits intake shortly before the appointment?
9. Which notification events are mandatory, optional, or clinic-configurable?
10. What level of operational tooling is required for staff to retry failed synchronization attempts?

12. Recommended Modeling Scope

For the initial architecture model, focus on:

1. Appointment scheduling and appointment management.
2. Intake submission and review.
3. Reminder delivery.

4. Electronic health record synchronization.
5. Audit and integration status tracking.

Defer advanced modeling for:

1. Insurance eligibility.
2. Bidirectional EHR reconciliation.
3. Patient profile management.
4. Waitlist optimization.
5. Advanced analytics.